

Statistics Essentials for Data Science

Describing data, probability, distributions, inference and correlation

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Introduction

Statistics is the science of collecting, describing and interpreting data under uncertainty. It gives data scientists the tools to summarize large datasets, judge whether a pattern is real or just noise, and quantify how confident they can be in a conclusion. Nearly every method in data science, from A/B testing to machine learning, rests on statistical ideas.

This guide covers the essentials in the order they build on one another: descriptive statistics, probability, common distributions, sampling and the central limit theorem, confidence intervals, hypothesis testing, and correlation and regression. Short Python snippets using NumPy, SciPy and Pandas illustrate each idea.

Two branches of statistics

Branch	Goal	Example
Descriptive	Summarize the data you have	Average order value last month was 42.30
Inferential	Draw conclusions about a larger population from a sample	Is the new checkout page truly better than the old one?

Population versus sample

A **population** is the complete group you care about (all customers), while a **sample** is the subset you actually observe. Numbers describing a population are called **parameters** (mean μ , standard deviation σ); numbers computed from a sample are **statistics** ($\bar{x}$, s). Inference uses statistics to estimate parameters.

Types of data

- **Numerical:** continuous (height) or discrete (number of visits).
- **Categorical:** nominal (color) or ordinal (rating: low, medium, high).

```
import numpy as np, pandas as pd
from scipy import stats
```

1. Descriptive Statistics

Descriptive statistics condense a dataset into a few informative numbers. Two families matter most: measures of center and measures of spread.

Measures of center

- **Mean:** the sum of values divided by their count. Sensitive to outliers.
- **Median:** the middle value when sorted. Robust to outliers; best for skewed data such as income.
- **Mode:** the most frequent value; the only center that works for categories.

Measures of spread

- **Range:** maximum minus minimum.
- **Variance:** the average squared distance from the mean.
- **Standard deviation (SD):** the square root of variance, in the same units as the data.
- **Interquartile range (IQR):** Q3 minus Q1; the spread of the middle 50%.

When computing from a sample, divide the sum of squared deviations by $n - 1$ (Bessel's correction) rather than n , to avoid underestimating the population variance.

```
data = np.array([12, 15, 14, 10, 18, 95])      # note the outlier 95
print(np.mean(data))                          # 27.33 (pulled up by the outlier)
print(np.median(data))                       # 14.5 (barely affected)
print(np.std(data, ddof=1))                  # sample standard deviation
q1, q3 = np.percentile(data, [25, 75])
print(q3 - q1)                               # IQR
```

Shape of a distribution

- **Skewness:** a long right tail gives positive skew (mean greater than median).
- **Kurtosis:** how heavy the tails are relative to a normal curve.

Visualizing distributions

Chart	Shows
Histogram	Frequency of values in bins; overall shape
Box plot	Median, quartiles and outliers at a glance
Bar chart	Counts or values per category
Scatter plot	Relationship between two numeric variables

Rule of thumb: Always look at a plot before trusting a summary number. Very different datasets can share the same mean and standard deviation.

2. Probability Fundamentals

Probability measures how likely an event is, on a scale from 0 (impossible) to 1 (certain). It is the mathematical language for uncertainty and underpins all of inferential statistics.

Basic rules

- **Complement:** $P(\text{not } A) = 1 - P(A)$.
- **Addition (either A or B):** $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.
- **Multiplication (both A and B):** if independent, $P(A \text{ and } B) = P(A) \times P(B)$.

Example: the probability of rolling a six on a fair die is $1/6$. The probability of two sixes in a row is $1/6 \times 1/6 = 1/36$, because the rolls are independent.

Conditional probability

$P(A | B)$ is the probability of A given that B has occurred: $P(A | B) = P(A \text{ and } B) / P(B)$. Conditioning changes the sample space, and it is central to machine learning and medical testing.

Bayes' theorem

Bayes' theorem reverses a conditional probability: $P(A | B) = P(B | A) \times P(A) / P(B)$. It shows how to update beliefs when new evidence arrives.

Worked example: a disease affects 1% of people. A test detects it 95% of the time when present, and gives a false alarm 5% of the time when absent. If someone tests positive, what is the chance they are sick?

```
p_d = 0.01                # prevalence
p_pos_d = 0.95            # sensitivity
p_pos_not = 0.05          # false positive rate

p_pos = p_pos_d * p_d + p_pos_not * (1 - p_d)
posterior = p_pos_d * p_d / p_pos
print(round(posterior, 3))  # 0.161 -> only about 16%
```

Despite a 95% accurate test, the chance of disease is only about 16% because the condition is rare. Ignoring the base rate is one of the most common reasoning errors.

Expected value

The expected value is the long-run average of a random outcome, computed as the sum of each value times its probability. A game paying 10 with probability 0.2 and 0 otherwise has an expected value of 2.

Key points

- Probabilities are between 0 and 1 and the outcomes of an experiment sum to 1.
- Independence means one event does not change the probability of another.
- Always consider base rates when interpreting test results.

3. Common Probability Distributions

A probability distribution describes how the possible values of a random variable are spread out. A handful of distributions cover most practical situations.

Distribution	Type	Models	Example
Bernoulli / Binomial	Discrete	Number of successes in n yes/no trials	Clicks among 100 visitors
Poisson	Discrete	Count of events in a fixed interval	Support tickets per hour
Uniform	Continuous	All values equally likely	Random number generator
Normal (Gaussian)	Continuous	Bell-shaped natural variation	Heights, measurement errors
Exponential	Continuous	Waiting time between events	Time until next customer

The normal distribution

The normal distribution is symmetric and fully described by its mean and standard deviation. It appears everywhere because of the central limit theorem (next section). The **empirical rule** states that about 68% of values fall within 1 SD of the mean, 95% within 2 SD, and 99.7% within 3 SD.

Z-scores

A z-score expresses how many standard deviations a value lies from the mean: $z = (x - \text{mean}) / \text{SD}$. It allows comparison across different scales and flags unusual values (commonly $|z|$ greater than 3).

```
from scipy import stats

# Exam scores ~ Normal(mean=70, sd=10). Probability of scoring above 85?
print(1 - stats.norm.cdf(85, loc=70, scale=10))    # 0.0668

# z-score of 85
print((85 - 70) / 10)                              # 1.5

# Binomial: probability of exactly 7 heads in 10 fair coin flips
print(stats.binom.pmf(7, n=10, p=0.5))            # 0.1172
```

Key points

- Choose a distribution based on the process that generates the data.
- **pmf** gives probabilities for discrete values; **pdf** gives density for continuous ones; **cdf** gives the cumulative probability.
- Z-scores standardize values to a common scale.

4. Sampling and the Central Limit Theorem

We rarely observe an entire population, so we take a sample. How the sample is chosen determines whether conclusions can be trusted.

Sampling methods

- **Simple random sampling:** every member has an equal chance of selection.
- **Stratified sampling:** divide the population into groups (strata) and sample from each.
- **Cluster sampling:** randomly select whole groups, such as schools.
- **Convenience sampling:** whoever is easiest to reach; prone to bias and best avoided.

Bias warning: A large sample cannot fix a biased method. Surveying only online users to estimate national opinion is biased no matter how many people respond.

Sampling distributions and standard error

If you drew many samples and computed the mean of each, those means would form a **sampling distribution**. Its spread is the **standard error**: $SE = s / \sqrt{n}$. Larger samples give smaller standard errors and therefore more precise estimates, but quadrupling n only halves the error.

The central limit theorem (CLT)

The CLT states that, for a sufficiently large sample size (often n of 30 or more), the distribution of the sample mean is approximately normal, regardless of the shape of the original population, with mean equal to the population mean and spread equal to the standard error. It explains why the normal distribution and its associated tools are so broadly useful.

```
rng = np.random.default_rng(0)
population = rng.exponential(scale=10, size=100_000)    # strongly skewed

sample_means = [rng.choice(population, 50).mean() for _ in range(2000)]
print(np.mean(sample_means))    # close to the population mean (about 10)
print(np.std(sample_means))     # close to 10 / sqrt(50) = 1.41
# a histogram of sample_means looks bell-shaped even though the data is skewed
```

Key points

- Random sampling protects against bias; size improves precision.
- Standard error measures uncertainty in an estimate.
- The CLT lets us apply normal-based inference to means.

5. Confidence Intervals and Hypothesis Testing

Inference answers two related questions: how large is an effect (estimation), and is the effect real (testing)?

Confidence intervals

A **confidence interval (CI)** gives a range of plausible values for a population parameter. A 95% CI is built so that, if the procedure were repeated many times, about 95% of the intervals would contain the true value. For a mean, it is approximately: sample mean plus or minus $t \times SE$.

```
sample = np.array([52, 48, 55, 60, 47, 51, 58, 49, 53, 56])
mean, se = sample.mean(), stats.sem(sample)
ci = stats.t.interval(0.95, df=len(sample) - 1, loc=mean, scale=se)
print(mean, ci)          # 52.9 (49.8, 56.0) approximately
```

The logic of hypothesis testing

- 1 State the **null hypothesis (H0)**: no effect or no difference.
- 2 State the **alternative hypothesis (H1)**: the effect exists.
- 3 Choose a significance level **alpha** (commonly 0.05).
- 4 Compute a **test statistic** and its **p-value**.
- 5 If p is less than alpha, reject H0; otherwise, fail to reject it.

The **p-value** is the probability of seeing data at least as extreme as yours if the null hypothesis were true. It is not the probability that the null hypothesis is true.

Common tests

Test	Use when	SciPy function
One-sample t-test	Compare a mean to a known value	<code>ttest_1samp</code>
Two-sample t-test	Compare the means of two groups	<code>ttest_ind</code>
Paired t-test	Before/after on the same subjects	<code>ttest_rel</code>
Chi-square test	Association between two categorical variables	<code>chi2_contingency</code>
ANOVA	Compare means of three or more groups	<code>f_oneway</code>

```
# A/B test: does the new page yield longer sessions?
a = [3.1, 2.8, 3.5, 3.0, 2.9, 3.3, 3.2, 2.7]
b = [3.6, 3.9, 3.4, 4.0, 3.7, 3.5, 3.8, 3.3]
t, p = stats.ttest_ind(a, b, equal_var=False)
print(t, p)          # p is well below 0.05 -> evidence of a difference
```

Errors and power

- **Type I error:** rejecting a true H0 (false positive); its probability is alpha.
- **Type II error:** failing to reject a false H0 (false negative).

- **Power:** the probability of detecting a real effect; increases with larger samples and larger effects.

Beware: Statistical significance is not practical importance. A tiny effect can be significant with huge data. Always report effect size and a confidence interval, and avoid testing many hypotheses without correction (p-hacking).

6. Correlation, Regression and Common Pitfalls

Many questions concern the relationship between variables. Correlation measures its strength; regression models it.

Correlation

The Pearson correlation coefficient r ranges from -1 (perfect negative linear relationship) through 0 (none) to +1 (perfect positive). Spearman's rank correlation handles monotonic but non-linear relationships and is robust to outliers.

```
hours = np.array([1, 2, 3, 4, 5, 6, 7, 8])
score = np.array([52, 55, 61, 64, 70, 74, 79, 85])
r, p = stats.pearsonr(hours, score)
print(round(r, 3), p)      # r is about 0.998, strongly positive
```

Correlation is not causation: Ice cream sales and drownings rise together because both increase in summer. A hidden third factor (a confounder) can create a relationship without any direct link. Only well-designed experiments, such as randomized trials, can establish causation.

Simple linear regression

Regression fits a line $y = b_0 + b_1 * x$ that minimizes squared errors. The slope b_1 estimates how much y changes for a one-unit increase in x . R-squared reports the share of variance explained.

```
res = stats.linregress(hours, score)
print(res.slope, res.intercept)      # about 4.7 points per extra hour
print(res.rvalue ** 2)               # R-squared
print(res.slope * 9 + res.intercept) # prediction for 9 hours
```

Regression assumes a roughly linear relationship, independent errors with constant spread, and no severe outliers. Inspect residual plots to check.

Common pitfalls

- **Simpson's paradox:** a trend within groups can reverse when groups are combined.
- **Survivorship bias:** analyzing only the cases that made it through a filter.
- **Overfitting to noise:** finding patterns in random data through too many comparisons.
- **Extrapolation:** predicting far beyond the range of observed data.

Conclusion

Statistics provides a disciplined way to move from raw numbers to reliable conclusions. You have reviewed how to describe data with measures of center and spread, reason about uncertainty with probability, model variation with common distributions, understand sampling and the central limit theorem, quantify precision with confidence intervals, test claims with hypothesis tests, and study relationships with correlation and regression.

Key takeaways

- Summarize with both a center and a spread, and always plot the data.
- Consider base rates before interpreting probabilities (Bayes' theorem).
- Larger, unbiased samples produce more precise estimates; the standard error shrinks with the square root of n .
- A p-value measures compatibility with the null hypothesis, not the probability that it is true.
- Correlation does not imply causation; report effect sizes and intervals, not only significance.

Suggested next steps

- 1 Practice with Pandas and SciPy on real datasets, computing intervals and running tests.
- 2 Learn multiple regression, logistic regression and regularization.
- 3 Study experimental design, A/B testing and power analysis.
- 4 Explore Bayesian statistics for a probabilistic view of inference.

Remember: The goal of statistics is not to produce a p-value but to make better decisions under uncertainty. Ask what could be wrong with the data before trusting the answer.